

Disproof of conjecture 00000000421:
the expected number of addable cells is at least $7/3$ at every prime
size

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1 The conjecture and the reading

Statement (English, verbatim). Definition: The growth-tree statistic $a(T)$ is the number of outer-corner cells that can be added to the tableau T . Conjecture: The expectation of $a(T)$ on a uniformly random tableau converges to two; and the convergence rate is the inverse logarithm of the tableau size. (growth-tree expectation limit two) (The Chinese text states the same.)

Reading. The growth tree is Young's lattice: a tableau grows by adding one cell at a time, and the cells that can be added to T are the addable (outer-corner) cells of its shape. For $n \geq 1$ let E_n be the expectation of $a(T)$ when T is a uniformly random standard Young tableau with n cells. The conjecture asserts

$$(L) \quad E_n \rightarrow 2 \quad (n \rightarrow \infty) \quad \text{and} \quad (R) \quad |E_n - 2| \text{ is of order } 1/\log n.$$

Every version of (R) (namely $|E_n - 2| = O(1/\log n)$, $\Theta(1/\log n)$, or $E_n - 2 \sim c/\log n$) implies $|E_n - 2| \leq C/\log n$ for some constant C and all large n . We refute (L), and we refute this weakest form of (R). We also refute both clauses for a second reading of “tableau”, namely a Young diagram (Ferrers shape) with n cells chosen uniformly.

Readings not covered. (i) Semistandard tableaux: with unbounded entries there are infinitely many of each shape, so there is no uniform measure, and we do not treat bounded-entry variants. (ii) Counting removable (inner) corners instead of addable ones: the text says cells that “can be added”, so we do not treat this. (iii) The Plancherel measure $f_\lambda^2/n!$: the counting argument below applies to it as well (the row and the column have weight $1/n!$ each), but this is not formalized.

2 Definitions (as in the Lean file)

Definition 1. A *Young diagram* is Mathlib's `YoungDiagram`: a finite set $\mu \subset \mathbb{N} \times \mathbb{N}$ that is a lower set for the product order. Cell (i, j) lies in row i and column j , counted from 0. We write $r = \text{rowLen } 0$ and $c = \text{colLen } 0$ for the lengths of the first row and the first column. Mathlib proves $(i, j) \in \mu \iff j < \text{rowLen } i \iff i < \text{colLen } j$.

Definition 2 (addable cells, `addable`, `a`). A cell $x \notin \mu$ is *addable* (an outer corner) if $\mu \cup \{x\}$ is again a lower set, that is, a Young diagram. The statistic is $a(\mu) = \#\{\text{addable cells of } \mu\}$, written in Lean as `Set.ncard`. We prove that this set is finite (`addable_finite`).

Definition 3 (standard Young tableaux, SYT). A standard Young tableau with n cells is recorded by the injective map $\text{pos} : \{0, \dots, n-1\} \rightarrow \mathbb{N} \times \mathbb{N}$, where $\text{pos}(k)$ is the cell holding the entry $k+1$. Its image is required to be a lower set (the shape SYT.shape , a Young diagram with n cells), and entries must increase from left to right along each row and from top to bottom down each column: if $\text{pos}(k)$ and $\text{pos}(l)$ are in the same row (column) and $\text{pos}(k)$ is to the left of (above) $\text{pos}(l)$, then $k < l$. We prove that there are finitely many such tableaux, since all coordinates are $< n$ (SYT.pos_lt). The statistic of a tableau is a of its shape.

Definition 4 (the two expectations). $E_{\text{SYT}}(n) = \frac{1}{\#\text{SYT}(n)} \sum_{T \in \text{SYT}(n)} a(\text{shape } T)$ (E_SYT), and $E_{\text{shape}}(n) = \frac{1}{\#\text{Shape}(n)} \sum_{\mu \in \text{Shape}(n)} a(\mu)$ (E_shape), where $\text{Shape}(n)$ is the set of Young diagrams with n cells.

3 Proof

Lemma 5 (`corner_row`, `corner_col`, `two_le_a`). *The cells $(0, r)$ and $(c, 0)$ are addable. If $\mu \neq \emptyset$, they are distinct, so $a(\mu) \geq 2$.*

Proof. A cell $(i, j) \notin \mu$ is addable as soon as $(i-1, j) \in \mu$ when $i > 0$ and $(i, j-1) \in \mu$ when $j > 0$ (`mem_addable_of`). Indeed, any $y < (i, j)$ in the product order is below one of these two cells, so it lies in μ . Now $(0, r) \notin \mu$, and if $r > 0$ then $(0, r-1) \in \mu$; if $r = 0$ the condition is vacuous (for the empty diagram $(0, r) = (0, 0)$ is addable). The case of $(c, 0)$ is symmetric, using $c > 0$ in the same way. If $(0, 0) \in \mu$, then $r, c \geq 1$, so $(0, r) \neq (c, 0)$. $\square$

Lemma 6 (`three_le_a`). *If some cell (i, j) with $i < c$ and $j < r$ is not in μ (so μ is not the $c \times r$ rectangle), then $a(\mu) \geq 3$.*

Proof. Let k be the least row with `rowLen` $k < r$. Such a row exists, since `rowLen` $i \leq j < r$, and $k \leq i < c$. Also $k \geq 1$, because row 0 has length r . Since $k < c$ we have $(k, 0) \in \mu$, so `rowLen` $k \geq 1$. The cell $(k, \text{rowLen } k)$ is not in μ , and the cell to its left is in μ . The cell above it, $(k-1, \text{rowLen } k)$, is also in μ , because `rowLen` $(k-1) \geq r > \text{rowLen } k$ by minimality of k . So it is addable. It differs from $(0, r)$ because $k \geq 1$, and from $(c, 0)$ because $k < c$. $\square$

Lemma 7 (`rowOrCol_or_three_le`). *If $\#\mu = p$ is prime, then μ is a single row ($\forall x \in \mu, x_1 = 0$), or a single column ($\forall x \in \mu, x_2 = 0$), or $a(\mu) \geq 3$.*

Proof. If the hypothesis of the previous lemma fails, every cell of the box $[0, c) \times [0, r)$ is in μ . Conversely, each cell $(i, j) \in \mu$ has $i < c$ and $j < r$. So μ is that box and $p = c \cdot r$. Since p is prime, $c = 1$ (a row) or $r = 1$ (a column). $\square$

Lemma 8 (`SYT.row_unique`, `SYT.col_unique`, `shape_eq_of_row`, `shape_eq_of_col`). *There is at most one standard tableau with n cells whose shape is a single row, and at most one whose shape is a single column. Two single-row (or two single-column) diagrams with the same number of cells are equal.*

Proof. For a single-row tableau, the row condition and injectivity make $k \mapsto \text{pos}(k)_2$ a strictly increasing map $\{0, \dots, n-1\} \rightarrow \{0, \dots, n-1\}$. Mathlib's `Finset.orderEmbOfFin_unique` shows that such a map is unique, so any two single-row tableaux coincide. Columns are handled in the same way. A single-row diagram with n cells is $\{(0, j) : j < n\}$, and the same holds for a single column. $\square$

Lemma 9 (`hook, hook_not_rowOrCol`). For $n \geq 3$, placing 1 at $(0,0)$, 2 at $(1,0)$, and $3, \dots, n$ at $(0,1), \dots, (0, n-2)$ gives a standard tableau. Its shape (the hook $(n-1,1)$) is neither a row nor a column.

Lemma 10 (`avg_bound`). Let X be a finite set with a map $\text{sh} : X \rightarrow \{\text{Young diagrams}\}$. Assume that every $\text{sh}(x)$ has a prime number of cells, that at most one x has single-row shape and at most one has single-column shape, and that some x_0 has another shape. Then $\frac{1}{\#X} \sum_x a(\text{sh } x) \geq 7/3$.

Proof. Let B be the set of x with row or column shape, so $|B| \leq 2$ and $|B| + 1 \leq t := \#X$. Lemma 1 (the shapes are nonempty) and Lemma 3 give $a \geq 3 - [x \in B]$ pointwise, so $\sum_x a \geq 3t - |B|$. Since $3|B| \leq 2|B| + 2 \leq 2t$, we get $3t - |B| \geq \frac{7}{3}t$. $\square$

Theorem 11 (`E_SYT_ge, E_shape_ge`). For every prime $p \geq 3$, $E_{\text{SYT}}(p) \geq 7/3$ and $E_{\text{shape}}(p) \geq 7/3$.

Proof. Apply the previous lemma with $X = \text{SYT}(p)$ and $\text{sh} = \text{shape}$, and with $X = \text{Shape}(p)$ and $\text{sh} = \text{id}$, taking x_0 to be the hook tableau or its shape. $\square$

Theorem 12 (`not_limit_two, disproof`). Let E be E_{SYT} or E_{shape} . Then $E(n) \not\rightarrow 2$. Moreover, there is no constant C such that $|E(n) - 2| \leq C/\log n$ for all large n .

Proof. Either statement would give $E(n) < 7/3$ for all large n . For (R), this uses $C/\log n \rightarrow 0$. But there are arbitrarily large primes (`Nat.exists_infinite_primes`), and at each of them $E \geq 7/3$. $\square$

Remark 13 (not used, not formalized). In fact $E_{\text{SYT}}(n) = t_{n+1}/t_n$, where t_n is the number of standard tableaux with n cells: deleting the largest entry is a bijection from tableaux with $n+1$ cells onto pairs (tableau with n cells, addable cell). OEIS A000085 (<https://oeis.org/A000085>) identifies t_n : “Number of self-inverse permutations on n letters, also known as involutions; number of standard Young tableaux with n cells.” It also gives the recurrence “ $a(n) = a(n-1) + (n-1)a(n-2)$ ”. Hence $E_{\text{SYT}}(n) \rightarrow \infty$. Our script checks $E_{\text{SYT}}(n) = t_{n+1}/t_n$ numerically for $n \leq 22$.

4 Lean formalization and axiom audit

The file `lean/Conjecture421/Basic.lean` (Lean 4, Mathlib v4.33.1; its only import is `Mathlib`) proves

```
theorem C421.disproof :
  (forall p : Nat, p.Prime -> 3 <= p -> 7/3 <= E_SYT p /\ 7/3 <= E_shape p) /\
  (not Tendsto E_SYT atTop (nhds 2) /\
   not exists C : Real,
    eventually (n in atTop), |E_SYT n - 2| <= C / Real.log n) /\
  (not Tendsto E_shape atTop (nhds 2) /\
   not exists C : Real,
    eventually (n in atTop), |E_shape n - 2| <= C / Real.log n)
```

(written here in ASCII; the source uses the usual Unicode notation). All the lemmas above appear under the names given. It uses Mathlib’s `YoungDiagram` API (`rowLen`, `colLen`, `mem_iff_lt_rowLen`, `mem_iff_lt_colLen`, `rowLen_eq_card`, `colLen_eq_card`, `up_left_mem`). No code was taken from other submissions. `lake build` succeeds, and `#print axioms C421.disproof` reports only `[propext, Classical.choice, Quot.sound]`. The file contains no `sorry`, no `native_decide` and no added axioms.